

1. E-Commerce Security

E-commerce security means keeping online payments and customer data safe from hackers. Here are the most important rules to follow. We use simple words so everyone can understand.

Protect Data & Use Encryption

- ✓ **Follow PCI DSS(Payment Card Industry Data Security Standard) rules.** This is a set of rules for anyone who handles credit card data. Never store raw card numbers, CVV codes, or full card details unless you are fully allowed and protected.
- ✓ **Use HTTPS everywhere.** This locks the connection between the customer and your website. It stops hackers from reading data while it travels. Use TLS 1.3 or newer.
- ✓ **Tokenization.** Replace real card numbers with random tokens. If a hacker steals a token, it is useless outside your system.
- ✓ **Encrypt stored data.** Use strong encryption (like AES-256) for databases and backups. Keep encryption keys safe using special hardware or cloud key managers.
- ✓ **Collect only what you need.** Do not ask for extra personal data. Follow privacy laws like GDPR and CCPA. Delete old data you no longer need.

Control Who Can Access & Verify Identity

- ✓ **Use multi-factor authentication (MFA).** This means asking for two or more proofs of identity. For example: a password plus a code from your phone. Use it for

admin panels and customer accounts.

- ✓ **Give the least access possible.** Employees should only see data they truly need. This is called "least privilege" or RBAC (role-based access control).
- ✓ **Use 3-D Secure.** This adds an extra check for online card payments. The bank may ask for a password or fingerprint. It lowers fraud.
- ✓ **Watch for fraud.** Use simple checks: is the address correct? Is the CVV right? Is the same card being used too many times too fast? Use machine learning to spot strange activity.
- ✓ **Keep APIs safe.** Use OAuth 2.0, API keys with limited power, and check the signature on webhooks. Never put secret keys in the browser code.

Keep Systems Strong

- **Update everything.** Patch your website, plugins, server, and libraries. Old software has holes that hackers love.
- **Use a firewall (WAF).** It blocks bad traffic like SQL injection and DDoS attacks.
- **Write safe code.** Check all input, clean all output, and never hardcode passwords.
- **Test your security.** Do regular penetration tests and vulnerability scans.
- **Watch your logs.** Keep records of activity and set up alerts for strange events.

Follow Rules & Privacy Laws

- **PCI DSS** — Must follow if you touch card data.
- **GDPR / CCPA** — Protect user privacy and allow data deletion.
- **PSD2 / SCA** — Strong customer authentication for Europe.
- **AML / KYC** — Check who your customers are to stop money laundering.

- **Have a breach plan.** Know what to do if data is stolen.



Remember: Security is not a one-time job. Hackers change their tricks every day. You must check your systems often, train your staff, and stay alert. One small hole can let a hacker into everything.



2. What Are Digital Payments?

Digital payments are payments made using computers, phones, or the internet. No physical cash is needed. Money moves from one person or business to another electronically. This is how most online shopping works today.



Main Features



Very fast — often in seconds



Works across countries



Works on phones and contactless



Often automatic



Gives data for analysis



Common Types

- Credit / debit cards
- Digital wallets (PayPal, Apple Pay)
- Bank transfers (ACH, SEPA, wire)
- Cryptocurrency
- Buy Now Pay Later
- QR codes (UPI, Alipay)
- Direct carrier billing
- Open banking payments

How a Payment Moves



1. Authorization

The buyer starts the payment. The bank checks if there is enough money and says yes or no. This may include a password, fingerprint, or PIN.



2. Clearing

The banks share details and calculate fees. They match up the records.

3. Settlement

The real money moves between banks. The seller gets paid, often in 1 or 2 days.



4. Chargeback or Refund

If the buyer complains, the money may be sent back. The seller must prove the sale was real.

Types of Digital Payments — Simple Comparison

Type	Example	How Fast?	What to Watch For
Card payments	Visa, Mastercard	1–3 days	PCI rules, 3-D Secure, tokenization
Digital wallets	PayPal, Apple Pay	Instant to 1 day	MFA, device checks, encryption
Bank transfers	ACH, SEPA, Wire	Same day to 3 days	Check account details, AML/KYC

Type	Example	How Fast?	What to Watch For
Real-time payments	UPI, RTP, FedNow	Seconds	Strong login, fraud scoring
Cryptocurrency	Bitcoin, USDC	Minutes to hours	Cannot be reversed, keep wallet safe

3. Parts of Digital Payment & People Involved

A digital payment is like a machine with many parts. Each part has a job. Also, many different people and companies are involved. Let's look at each one in simple words.



Main Parts (Components)



1. Buyer & Seller

The two main people. The buyer pays. The seller receives. Both need to prove who they are.



2. Payment Tool

What the buyer uses to pay: a card, a wallet, a bank account, or crypto. This must be kept safe and secret.



3. Payment Network

The roads that money travels on. Examples: Visa, Mastercard, ACH, SWIFT, blockchain. Each network has its own safety rules.



4. Payment Gateway & Processor

The middleman that says yes or no to a payment. Examples: Stripe, PayPal, Adyen. They also do encryption and fraud checks.



5. Banks (Acquiring & Issuing)

The buyer's bank (issuer) gives the card. The seller's bank (acquirer) receives the money. Both check for fraud.



6. Security & Rules Layer

Tokenization, encryption, 3-D Secure, PCI DSS, fraud detection, KYC/AML. This wraps around everything to keep it safe.



People & Groups Involved (Stakeholders)

Each person or group has a different job and a different security duty.



Customer / Buyer



Merchant / Seller



Issuing Bank

Acquiring Bank



Card Network



Payment Processor



Gateway Provider



Regulators (PCI, RBI, etc.)



Fraud & Risk Team



Developers / API Builders



How They Work Together

- **Customer ↔ Seller** – choose item and pay.
- **Seller ↔ Gateway** – send payment for approval.
- **Gateway ↔ Acquiring Bank** – settle the money.
- **Issuer ↔ Network** – check if buyer has funds.
- **Regulators ↔ Everyone** – make sure rules are followed.

 **Trust triangle:** Buyer, Seller, and Banks must all trust each other. Security makes that trust possible.



Simple summary: E-commerce security is about keeping data safe with encryption, strong passwords, and constant checking. Digital payments move money electronically through many parts. Everyone — from the customer to the bank — has a role in keeping payments safe. Follow the best practices, and you will build trust with your users.